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Review

Environmental pollution of electronic waste recycling in India: A critical review

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ABSTRACT

The rapid growth of the production of electrical and electronic products has meant an equally rapid growth in the amount of electronic waste (e-waste), much of which is illegally imported to India, for disposal presenting a serious environmental challenge. The environmental impact during e-waste recycling was investigated and metal as well as other pollutants [e.g. polybrominated diphenyl ethers (PBDEs), polychlorinated biphenyls (PCBs)] were found in excessive levels in soil, water and other habitats. The most e-waste is dealt with as general or crudely often by open burning, acid baths, with recovery of only a few materials of value. As resulted of these process; dioxins, furans, and heavy metals are released and harmful to the surrounding environment, engaged workers, and also residents inhabiting near the sites. The informal e-waste sectors are growing rapidly in the developing countries over than in the developed countries because of cheapest labor cost and week legislations systems. It has been confirmed that contaminates are moving through the food chain via root plant translocation system, to the human body thereby threatening human health. We have suggested some possible solution toward in which plants and microbes combine to remediate highly contaminated sites.

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1. Introduction

With rapid global advancement and an exponential growth rate in the electrical and electronic industries in the 21st century has come a corresponding change in consumer lifestyles, resulting in the generation of a huge amount of end-of-life electronics, known as electronic waste (e-waste) (Kiddee et al., 2013a, 2013b; Li et al., 2015a,b; Perez-Belis et al., 2015; Wang et al., 2013). It has been estimated that approximately 42 million tons (Mt) of e-waste is generated globally per annum (Balde et al., 2015).

About 80% of e-waste from developed countries is illegally exported to developing countries especially China, India, Nigeria, Ghana and Pakistan, because of the lower labor costs and lack of governmental regulations (Sthiannopkao and Wong, 2013; UNEP, 2005). According to Rajya Sabha's report (2011), almost all e-waste in India is collected and recycled in the informal sector, which has led to serious environmental problems (Keller, 2006; Needhidasan et al., 2014). E-waste has a high content of heavy metals, such as lead and cadmium in circuit boards, cadmium in batteries, and copper for electrical wiring and large amounts of these valuable metals remain after the disposal of e-products (Stevels et al., 2013; Tang et al., 2010a, 2010b, 2010c; Zeng et al., 2013). However, Bart Gordon, who served as Chairman of the U.S. House Committee on Science and Technology from 2007 to 2011, suggested that electronics engineers be required to know about the prospective ecological, social and health effects of e-waste, and take this knowledge into account when designing new electronic products (Ogunseitan et al., 2009).

E-waste contains two major types of substances:— hazardous [(Cd, Cr, Pb, Hg, Chlorofluorocarbon, (PAHs), (PBDEs), (PCDD/Fs)] and non-hazardous (base metals such as Cu, Se, Zn and precious metals such as Ag, Au, and Pt) both types have potential negative environmental impacts (Tsydenova and Bengtsson, 2011; Widmer et al., 2005; Wei et al., 2014; Zeng, 2014; Zhang et al., 2013). In addition, many organic pollutants such as polyaromatic hydrocarbons (PAHs), polychlorinated biphenyl (PCBs), Brominated flame retardants (BFRs), Polybrominated diphenyl ethers (PBDEs) and polychlorinated dibenzo-p-dioxin furans (PCDD/Fs)) are released into the environment during improper e-waste processing (Kiddee et al., 2013b; Zhang et al., 2012). Its impacts have played a major role in ecological risk assessments show that the heavily contaminated soil by PAHs is concentrated in the densely populated soil

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and number of residents subjecting to high health risk (Wei et al., 2014).

These substances are extensively used in many electronic products. India is identified as a major dumping site for e-waste as raw materials from developed countries. Most of the e-waste in India is recycled in informal workshops that perform operations such as precious metals recovery and the extraction of repairable parts in whatever ways are quickest and easiest, regardless of environmental considerations. Backyard family workshops in particular often use primitive recycling methods with high health risks (Toxic Link, 2014; Sinha-Khetriwal et al., 2005; Streicher-Porte et al., 2005), and these e-waste recycling practices are carried out all over India (Toxic Link, 2014; Sepulveda et al., 2010; Sinha-Khetriwal et al., 2005).

These workshops proliferate because of the availability of extremely cheap labor and the lack of governmental regulation and oversight. A number of investigations have suggested that the natural environment (soil, air, water, plants, etc.) is contaminated by exposure to the toxic substances released at these workshop sites (Jain and Sareen, 2006; Kwatra et al., 2014; Pradhan and Kumar 2014; Stevels et al., 2013; Zeng et al., 2013; Fujimori and Takigami, 2014; Hites, 2004; Song and Li 2014a; 2014b; Wu et al., 2014). Many published studies have documented heavy metal contamination in the soil, air and water near recycling sites in developing countries (Leung et al., 2006, 2007; 2008; Sharma et al., 2007; Steiner, 2004; Wong et al., 2007c).

Though this informal e-waste recycling technology extracts valuable metals rapidly, the recovery is inefficient and incomplete (Achillas et al., 2013; Luo et al., 2011; Song and Li, 2014a). Furthermore, it generates waste water containing high levels of toxic metals (Cd, Cu, Ni, Pb, and Zn) and other pollutants, which are discharged into the local environment, causing soil, air, water, and plant pollution (Deng et al., 2008; Wong et al., 2007a, 2007b; Sojinu et al., 2012). A study in Bangalore, India suggested that informal processing of e-waste is responsible for heavy metals contamination in nearby soil and in human tissues, because of high penetration rates into the soil and thence into plants, where it accumulates and is consumed by humans (Ha et al., 2009). Zhao et al. (2010) and Grant et al. (2013) determined that there is a high probability of transfer of heavy metals and PBDEs from contaminated food plant to human beings, where they pose health hazards such as lung, liver and kidney damage (Chan et al., 2013; Li et al., 2011).

Greenpeace International's published report estimated that contamination from the recycling of electronic waste in China and India is 80% higher than that in the rest of the world (Brigden et al., 2005). These family workshop backyard often take place under the very primitive recycling methods with high health risk (Toxic Link, 2014; Sinha-Khetriwal et al., 2005; Streicher-Porte et al., 2005). This review compiled all the published literature related to the environmental assessment of metals such as chromium (Cr), cadmium (Cd), mercury (Hg), lead (Pb), as well as PBDEs and PCDD/Fs in e-waste processing sites in India. India is particularly vulnerable to these problems because it is one of the two countries most affected by improper recycling activities, as it has both an urgent need for material resources and a large number of people willing to work for very low wages. Keller (2006) highlighted some of the problems that was associated e-waste recycling in Bangalore, such as recovery of gold through chemical leaching processes. Due to improper processing, high levels of heavy metals and other pollutants were reported in areas such as New Delhi, Bangalore, Kolkata, Hyderabad, Trichirappalli and Gaziabad. Children are especially vulnerable to the harmful effects of these improper recycling activities. However, only limited studies have been carried out to explore the environmental degradation caused by heavy metals and other pollutants emitted from e-waste management or processing sites in India.

The objective of this study was to assess pollution levels and to provide comprehensive information on the impact of pollutants released from e-waste recycling sites into the natural environment. A detailed comparison of e-waste recycling and management facilities, and their expected impacts on natural environment was carried out for India. In addition China and Nigeria could be consider as further proof or reference in order to determine impact of informal e-waste recycling. Through this review, we explored the environmental pollution generated from e-waste recycling in India. Finally, this study strives to outline proposed eco-friendly solution that may be helpful for resolving the problem and can be recommend to the Ministry of Environment, Forestry and Climate Change (MoEFCC) India.

2. Review methodology

This study analyzes information collected from more than a hundred published researches covering several parts of India, focusing on the environmental impact of heavy metals and other pollutants from e-waste processing. This study is similar to the work of Song and Li (2014a), but includes a number of additional pollutants, and takes a more remedial approach to the problem. The studies reviewed here include those from peer-reviewed journals, technical reports, thesis reports and conference proceedings published up through the end of May 2015. Some additional manuscripts from the e-waste recycling portions of reviews by Song and Li (2014a) and Zheng et al. (2012) were also considered. The main aim of this study was to evaluate the e-waste recycling activities particularly in different selected cities of India. These selected cities are highly dominating by informal recycling activities because of huge numbers of peoples are migrated for employment into these cities and also availability of e-waste as livelihood option. According to available research papers, reports and news, the informal sectors are systematically organized in most of these cities of India. These informal sector conveniently provides services for collection, segregation, dismantling and recycling of e-waste. Interestingly such units are spread in small clusters around or all over these cities and almost 95% of e-waste recycling by these informal sector in India. We also searched several different databases such as Science Direct and Google Scholar, using the key words ‘e-waste’, ‘electronic waste’ and ‘WEEE’, as well as the names of various heavy metals and other known pollutants such as ‘PBDE’,’PCB’ and ‘organochlorien’ (see Fig. 1).

3. Analysis and discussion

3.1. Overview of cross-relationship between environmental medium and e-waste recycling process

Mainly three kind of substances released during recycling (a) The substances used in manufacturing of electrical and electronic equipment (b) Those substance are used in recycling process (auxiliary substances) (c) By products which are formed during the transformation of primary constituents (Sepulveda et al., 2010). The details diagrammatical representation of environmental medium and e-waste recycling shown in figure (Fig. 2).

The present systemic review analyzes the environmental pollution effects of heavy metals and other pollutants from e-waste processing workshops in India. We only evaluated the possible exposure routes and human health risk due to effects of heavy metals in order to understand the evidence of causality between exposure to heavy metals from e-waste and human health outcomes. According to Toxic Link report (2014) various metals

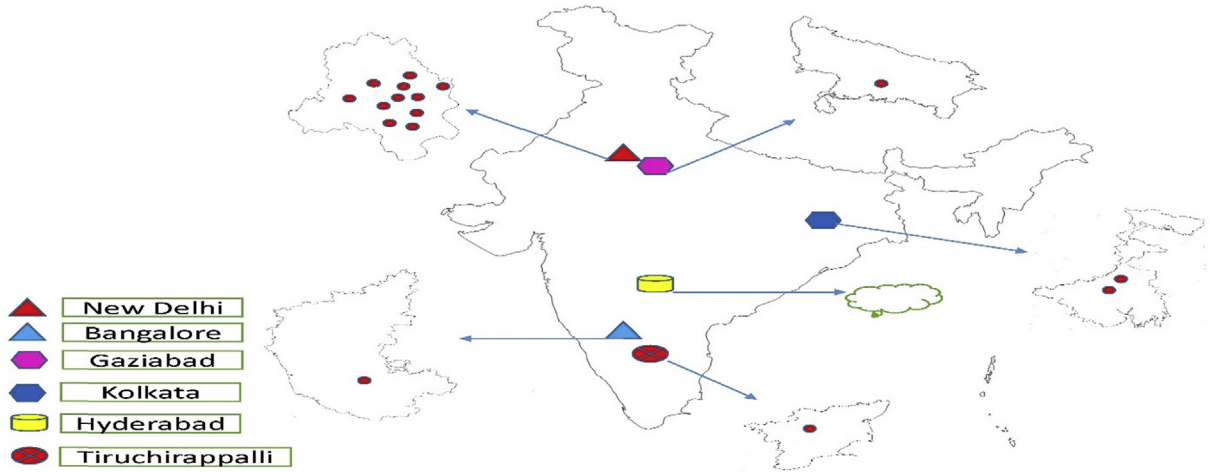


Fig. 1. Location of map with different cities and Impact of e-waste recycling in India.

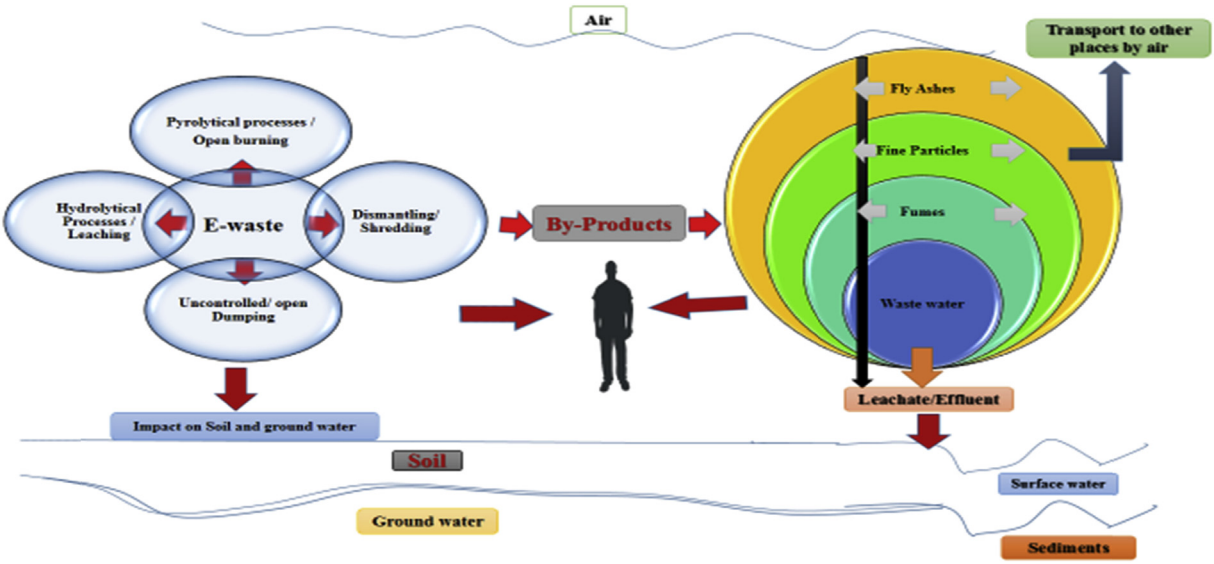


Fig. 2. Diagrammatical Illustration between environmental medium and e-waste recycling process.

including mercury, lead, and zinc are released during the e-waste recycling process and pollute nearby soil and water sources. Although many studies have suggested that e-waste recycling activities impact on natural environment (Orlins and Guan, 2015; Wang et al., 2011a).

This paper therefore explored the impacts of e-waste recycling practices on soil, air, water, vegetation, and other habitat elements near e-waste processing sites.

3.2. Heavy metal contaminates in dust and air

The dust samples collected from battery dismantling workshops in New Delhi were found to have high metal concentrations (Brigden et al., 2005). Table 1 shown the heavy metal concentration in dust and air reported from different regions of the country. The average level of heavy metals (Cr) in dust were, in decreasing order: Zarfarabad > Mayapuri > Brijgang (Dust storage shed) > Shastri Park (Sheperation workshop) > Shastri Park (Solder workshop) > Buradi > Kailash Nagar and Safouring > Brijgang (Ground storage shed) > Shastri park (Solder circuit Board) > Zarfarabad (soldered circuit board). Cadmium levels in dust ranged from 200,000 mg/kg in Buradi and Kailash Nagar to <0.5 mg/kg in Safouring. High concentrations of Pb—37,000 mg/kg and 20 mg/kg— were found in soldering workshops in Shastri Park and Safouring, respectively. A concentration of Cu above 6805 mg/kg was observed in dust in a separation workshop in Zarfarabad. Zn concentration levels ranged upto 21,100 mg/kg in Brijgang (dust storage shed) and <10 mg/kg in Zarfarabad (soldered circuit board). Levels of Hg ranged from a high of 48.2 mg/kg in Buradi (floor dust from a battery workshop) to a low of <0.2 mg/kg in Gaziabad, Shastri Park and Brijgang. Zhu et al. (2012) and Bi et al. (2011), reported high levels of metal concentration in dust from the e-waste recycling sector (Leung et al., 2008). Fang et al. (2013) found that the concentrations of Pb in workshop dust were higher than those from other studies (Leung et al., 2008). Song and Li (2014a) analyzed environmental pollution attributable to heavy metals from an e-waste processing site China. An investigation by Song et al. (2015a) found the highest concentration of heavy metals in the air and dust of a CRT workshop was of Pb—2.3 mg/m^3 and 10.53 mg/g in comparison with two other heavy metals studied: Cu

Table 1
Concentration of heavy metals in air and dust samples around e-waste recycling areas in India described in Brigden et al. (2005).

Location	Sampling site	Substance concentration (mg/kg)
Gaziabad, Utter Pradesh	Workshop floor dust	Cr: 20; Cu: 149; Zn: 549; Cd: 11.4; Hg: <0.2; Pb: 100; Mo: <2;
Zarfarabad, New Delhi	Solder, circuit board	Cr: <20; Cu: 2070; Zn: <10; Cd: <5; Hg: <10; Pb: 362,000; Mo: <20
Zarfarabad, New Delhi	floor dust, separation workshop	Cr: 158; Cu: 6805; Zn: 4440; Cd: 97; Hg: 460; Pb: 8815; Mo: 12
Shashtri Park, New Delhi	Solder, circuit board	Cr: <20; Cu: 2670; Zn: 21; Cd: <5; Hg: <10; Pb: 375,000; Mo: <20
Shashtri Park, New Delhi	Floor dust, separation workshop	Cr: 78; Cu: 2800; Zn: 2200; Cd: 14.1; Hg: 2.1; Pb: 2360; Mo: 7
Shashtri Park, New Delhi	Floor dust, solder workshop	Cr: 64; Cu: 2140; Zn: 1410; Cd: 15.5; Hg: 0.5; Pb: 10,900; Mo: 4
Shashtri Park, New Delhi	Street dust, workshop	Cr: 30; Cu: 230; Zn; 710; Cd: 1.4; Hg: <0.2; Pb: 48; Mo: <2
Mayapuri, New Delhi	Floor dust, Battery workshop	Cr: 103; Cu: 1730; Zn: 4920; Cd: 42.6; Hg: 3.5; Pb: 88,100; Mo: 7
Buradi, New Delhi	Floor dust, Battery workshop	Cr: 61; Cu: 1610; Zn: 1240; Cd: 200,000; Hg: 48.2; Pb: 13,300; Mo: 91
Kailash Nagar, New Delhi	Street dust, No workshop area	Cr: 25; Cu: 414; Zn: 414; Cd: <0.5; Hg: 0.6; Pb: 100; Mo: <2
Safourjung, New Delhi	Street dust, residential area	Cr: 25; Cu: 21; Zn: 83; Cd: <0.5; Hg: 0.5; Pb: 20; Mo: <2
Brijgang, New Delhi	Dust Storage shed	Cr: 86; Cu: 439; Zn: 21,100; Cd: 310; Hg: 0.5; Pb: 4600; Mo: <2
Brijgang, New Delhi	Dust, ground storage shed	Cr: 21; Cu: 82; Zn: 506; Cd: 16.4; Hg: <0.2; Pb: 1370; Mo: <2

and Cd.

Air is one of the most important source mediators for the survival and movement of e-waste dust pollutants released during recycling. Excess amounts of these pollutants, including heavy metals into the air contaminate the natural environment and inflict life-threatening effects on humans (Eguchi et al., 2012; Ejiogu, 2013; Bi et al., 2010). Deng et al. (2006) reported that, metals contaminating the air near e-waste recycling sites, the levels of Cr (1.161 lg/m^3) and Zn (1.038 lg/m^3) were the enriched metals with the highest level of total suspended particulates (TSP), higher than Cu (0.483 lg/m^3), Pb (0.444 lg/m^3), or Mn (0.0606 lg/m^3). In addition, a comparative study by Bi et al. (2010) found the concentration of heavy metals in formal recycling sector were lower than those in Guiyu, China. A study by Luo et al. (2011), found that the open burning of used electronic product released heavy smoke with a variety of both organic and heavy metals which contaminated the air.

3.3. Heavy metal contaminates in waste water (effluent), water and sediment

Waste water is another important transporter of contaminants (Liu et al., 2014; Wen et al., 2014; Sojinu et al., 2012; Yang et al., 2007). Table 2 shows the level of heavy metals in effluent, water and sediment attributable to improper e-waste processing. Several studies have examined heavy metal pollution from e-waste recycling sites (Green Cross, 2006). Nearby water is contaminated with heavy metals due to different acidification activity resulting from e-waste recycling activity on site. According to Pradhan and Kumar (2014), the levels of heavy metals in e-waste recycling area water were as follows: Cr (0.60 mg/l), Cu (0.70 (mg/l), Cd (0.05), Fe (0.46 mg/l), Pb (0.040 mg/l, Zn (1.89 mg/l), Al (3.67 mg/l) inside the unit, while levels found inside a residential area, 500 m away from the recycling site were at: Cr (0.02 mg/l), Cu (0.05 mg/l), Cd (0.002 mg/l), Fe (0.32 mg/l), Pb (0.002 mg/l), Zn (1.46 mg/l), Al (61 mg/l). Brigden et al. (2005) reported concentrations of Pb (0.06 mg/l) in groundwater samples; this is attributable to old technique and the fact that fewer measurements were taken in the e-waste workshop. Likewise, a Green Cross (2006) report concerning the level of heavy metals in water from an informal e-waste processing site in Kolkata found that the level of Fe (0.9 mg/l) in the pond adjacent to a picnic area had the highest levels of all the heavy metals, and lower than those of other investigations (Pradhan and Kumar 2014). These studies showed that New Delhi and Bangalore were changing improper e-waste processing in heavily polluted regions (Toxic Link, 2014). The formal e-waste recycling in India

Table 2
Concentration of heavy metals in water and sediments samples around e-waste recycling area in India described in previous reports.

Location	Sampling site	Substance concentration	References
Informal recycling site, Mandoli Delhi	Recycling site inside unit	Cr: 0.60; Cu: 0.70; Cd: 0.05; Fe: 0.46; Pb: 0.04; Zn: 1.89; Al: 3.67 mg/kg	Pradhan & Kumar, 2014
Informal recycling, Mandoli Delhi	Residential land, (500 m away from recycling site)	Cr: 0.02; Cu: 0.05; Cd: 0.002; Fe: 0.32; Pb: 0.002; Zn: 1.46; Al: 61 mg/kg	Pradhan & Kumar, 2014
	Groundwater in an area of separation of circuit boards and shredding	Pb: 0.06 mg/kg	Brigden et al., 2005
Azad Metal works, Kolkata India	Pond near Picnic garden (Pond Adjacent)	Fe: 0.9; Pb: 0.06; Zn: 0.07 mg/l	Green cross Report, 2006
Singh metal works, Eastern side	Pond Eastern side	Fe: 0.02; Pb: 0.03 mg/l	Green cross Report, 2006
Singh metal works, North side	Pond North side	Fe: 0.87; Pb: 0.05; Zn: 0.04 mg/l	Green cross Report, 2006
Mandoli, New Delhi	Waste water (waste acid solution)	Pb: 4 mg/l	Keller, 2006
	Wastewater acid processing site	Pb: 20.4 mg/l	Brigden et al., 2005
	Wastewater from separation of circuit boards and shredding	Pb: 46.9 mg/l	Brigden et al., 2005
	E-waste disposal site	Pb: 1.66; Cd: <0.011; Hg: <0.01; Cr: <0.01; Zn: 870; Ni: 1.36 mg/l	Toxic Link, 2014
Tiruchirappalli, India	Open dump site	Pb: 5.1485; Cu: 0.546; Cd: 1.0372 mg/l	Kanmani and Gandhimathi, 2013
Bangalore	Waste solution	Al: 1315; As: <0.5; Cd; <1; Cu: 185; Hg: <0.5; Ni: 9; Pb: 4; Zn: 17 ppm	Keller, 2006
Anand Metal Works	Pond Adjacent	Pb: 84; Cd: 0.01; Zn: 1.7 mg/kg	Green cross Report, 2006
Singh Metal Works	Pond on Eastern Side	Pb: 114; Cd: 0.01; Zn: 0.14 mg/kg	Green cross Report, 2006
Singh Metal Works	Ponds on North side	Pb: 95; Cd: 0.01; Zn: 0.09 mg/kg	Green cross Report, 2006

(Kwatra et al., 2014; Toxic Link, 2007) have comparatively high level of heavy-metal concentrations in the water, particularly Pb and Cu. Fang et al. (2013) reported that the Pb was flowing into the environment from recycling site, instead of Cr, Ni, Cu, and Cd, also added the mechanical process generated concentration of Cr, Cd and Ni, while manual dismantling of Cu and Pb respectively. Wong et al. (2007a) already proved that metal pollution concentrations of Cd, Cr, Cu, Mn, Ni, Pb and Zn were much higher at e-waste recycling sites.

3.4. Heavy metal contaminants in soil and ash

Heavy metal contamination in soil is a serious problem owing to its toxicity to both the environment and human health. Levels of heavy metals in the soil of e-waste recycling sites are shown in Table 3, and reveal significant differences between Mandoli and Bangalore. The average concentration of Cu was higher in the Mandoli industrial area (136,000 mg/kg), than in the formal recycling sector of Bangalore (22.8 mg/kg). Several researchers have already examined and reported that improper recycling of e-waste generates higher levels of metal (Lopez et al., 2011; Song and Li, 2014a). The concentration level of Cr metal in ash was observed to be highest in Ibrahimpur at 293 mg/kg, and lowest at 11 mg/kg in Kanti Nagar. The maximum concentration of Pb (20,500 mg/kg) and minimum (22.8 mg/kg). However, high level of heavy metals reported in soil of e-waste processing site in New Delhi. Brigden et al. (2005) suggested that the e-waste processing using primitive methods is a primary reason for soil contamination in India (Pradhan and Kumar 2014). The heavy metal profiles were different between all the studies. Ha et al. (2009) studied the impact of heavy metals contamination on surface soil in informal e-waste processing sites has damaged environmental quality (Guo et al., 2009; Orlins and Guan, 2015; Wong et al., 2007b). Similarly, the metal contamination surface soil near e-waste processing area was higher compared to uncontaminated soil (Pradhan and Kumar 2014). Man et al. (2010) explained about human health risks associated with soil contaminated by e-waste recycling process. According to Puckett et al. (2002), the potentially high concentration of heavy metals in soil is the result of the recycling technology used in processing sector. Li et al. (2011) and Luo et al. (2011) reported that the ash produced by burning e-waste contains high concentrations of heavy metals, such as Cu and Pb which could be deposited in the atmosphere (Bi et al., 2010; Gullett et al., 2007). Noel-Brune et al. (2013) determined that, there are two ways for e-waste workers or nearby residents to become contaminated: directly, during the recycling process; or indirectly, through the ecological cycle such as by intake of contaminated water (Wen et al., 2014) or via contaminated food chains. Excessive levels of e-waste contaminants inside the body leads to negative human health consequences (Robinson, 2009). The epidemiological work related to human e-waste exposure indicates that there are significant risks of respiratory, reproductive, genomic, and neurodevelopmental disorders (Grant et al., 2013; Ogunseitan, 2013; Zhang et al., 2012, 2013).

3.5. Heavy metal in contaminated plants

The root to shoot transportation of metal is a significant hyperaccumulater. The concentration of heavy metals in different plants is shown in Table 4. The previous studies revealed that

Table 3
Concentration of heavy metals in soil and ash samples around e-waste recycling area in India described in previous reports.

Location	Sampling site	Substance concentration	References
Formal recycling Bangalore, India	Slum site	Cr: 73; Cu: 592; Zn: 326; Cd: 2.33; Hg: 1.8; Pb: 297; Sb: 14; Mn: 449; V: 30; Co: 11; Mo: 1.78; Ag: 14; Sn: 86.1; TI: 0.39; BI: 0.66 ug/g	Ha et al., 2009
Formal recycling Bangalore, India	Recycling facility	Cr: 54; Cu: 429; Zn: 129; Cd: 0.47; Hg: 0.05; Pb: 126; Sb: 24; V: 53; Mn: 619; Co: 14; Mo: 1.8; Ag: 2.8	Ha et al., 2009
Formal recycling Bangalore, India	Control site	Cr: 57; Cu: 22.8; Zn: 41; Cd: 0.16; Hg: 0.05; Pb: 22.8; Sb; 0.43; V: 53; Mn: 390; Co: 11; Mo: 1.04; Ag: 0.36	Ha et al., 2009
Mandoli Industrial area Delhi	Burned fragment	Cr: 192; Cu: 136,000; Zn: 6400; Cd: 11.2; Hg: <0.2; Pb: 20,500 mg/kg	Brigden et al., 2005
Mandoli Industrial area Delhi	Ash from burning site	Cr: 103; Cu: 18,200; Zn: 2615; Cd: 6.7; Hg: 62.7; Pb: 3505 mg/kg	Brigden et al., 2005
Ibrahimpur, Delhi	Ash	Cr: 293; Cu: 13,500; Zn: 31,700; Cd: 66.6; Hg: 0.3; Pb: 3560 mg/kg	Brigden et al., 2005
Shashtri Park, Delhi	Ash	Cr: 54; Cu: 11,000; Zn: 897; Cd: 259; Hg: <0.2; Pb: 6350 mg/kg	Brigden et al., 2005
Kanti nagar, Delhi	CRT Powder, broken site	Cr: 11; Cu: 74; Zn: 273,000; Cd: 16,800; Hg: <0.2; Pb: 494 mg/kg	Brigden et al., 2005
Kanti nagar, Delhi	Soil ground under CRT	Cr: 20; Cu: 61; Zn: 964; Cd: 54.5; Hg: 0.3; Pb: 1580 mg/kg	Brigden et al., 2005
Brijgang, Delhi	CRT storage shed	Cr: 86; Cu: 439; Zn: 21,100; Cd: 310; Hg: 0.5; Pb: 14,600 mg/kg	Brigden et al., 2005
Informal recycling Mandoli Delhi	Recycling site inside unit	Ag: 12.38; Al: 8.822; As: 12.85; Cd: 1.14; Co: 13.25; Cu: 6734.8; Cr: 83.57; Fe: 4037.41; Hg: 0.07; Ni: 146.5; Pb: 2133.98; Se: 12.34; Zn: 416.31 mg/kg	Pradhan & Kumar, 2014
Informal recycling Mandoli Delhi	Dumping site	Ag: 10.75; Al: 14,142.58; As: 17.08; Cd: 1.29; Co: 12.43; Cu: 4291.61; Cr: 115.50; Fe: 4129.79; Hg: 0.08; Ni: 126.46; Pb: 2645.31; Se: 12.67; Zn: 776.84 mg/kg	Pradhan & Kumar, 2014
Informal recycling Mandoli Delhi	Arable land, (50 m away from recycling site)	Ag: 046; Al: 6476.44; As: 3.75; Cd: 0.70; Co: 4.94; Cu: 76.98; Cr: 34.79; Fe: 2952.25; Hg: nd; Ni: 44.67; Pb: 40.28; Se: 4.58; Zn: 90.28 mg/kg	Pradhan & Kumar, 2014
Informal recycling Mandoli Delhi	Arable land, (100 m away from recycling site)	Ag: 0.29; Al: 6538.78; As: nd; Cd: 0.06; Co: 2.32; Cu: 65.32; Cr: 22.00; Fe: 132.48; Hg: nd; Ni: 35.11; Pb: 29.61; Se: 4.23; Zn: 68.36 mg/kg	Pradhan & Kumar, 2014
Informal recycling Mandoli Delhi	Residential land, (500 m away from recycling site)	Ag: 0.29; Al: 6432; As: nd; Cd: 0.04; Co: 2.56; Cu: 63.45; Cr: 16.35; Fe: 1134.66; Hg: nd; Ni: 35.74; Pb: 27.94; Se: 4.57; Zn: 62.47 mg/kg	Pradhan & Kumar, 2014
Azad Metal works, Kolkata	Soil near Picnic	Cd: 0.01; Fe: 27; Pb: 84; Zn: 1.7 mg/kg	Green cross Report, 2006
Singh metal works, Eastern side Kolkata	Soil near Picnic garden	Cd: 0.01; Fe: 18; Pb: 114; Zn: 0.14 mg/kg	Green cross Report, 2006
Singh metal works, North side, Kolkata	Soil near Picnic garden	Cd: 0.01; Fe: 29; Pb: 95; Zn: 0.09 mg/kg	Green cross Report, 2006
Balanagar industrial area, Hyderabad	Dumping site	Cr: 4.41; Cu: 6.86; Zn: 12.02 mg/kg	Machender et al., 2010
Tiruchirappalli, India	Open Dumping site	Cu: 39.27; Cd: 30.58; Pb: 291.29 mg/kg	Kanmani, and Gandhimathi, 2013

dietary intake of plants containing residues of heavy metals exposes humans to metals and other pollutants. The food chain is an important pathway for the transportation of these contaminants into human beings (Ma et al., 2007). Singh et al. (2010) explained that the direct intake of toxic contaminants poses high health risks to humans (Bai et al., 2011). Plants growing in contaminated soil have the potential to accumulate heavy metal (Sharma et al., 2007; Singh et al., 2011). Palmgren et al. (2008) argued that this transport system's ability to accumulate toxic levels of pollutants such as heavy metals may be problematic in food crops. Sandalio et al. (2001) suggested that the heavy metals caused other negative impacts such as reduced plant growth (Di Salvatore et al., 2008; Lux et al., 2011). Plants are directly exposed to heavy metals as well as organic pollutants during the burning of e-waste. Bai et al. (2011) investigated the effect of heavy metals from e-waste recycling sites on agricultural and paddy soils from e-waste and found concentrations of Cu (663.08 mg/kg) and Cd (3.15 mg/kg) which exceeded standard limits. Garcia and Millan (1998) said that excessive heavy metals in plants can have negative impacts on food quality.

A plant root system growing in contaminated soil can accumulate significant quantities of metals with relative ease. Generally, plants obtain their mineral nutrient from the soil; depending on the species, plants can, to varying degrees, adapt to soils with high metal concentrations. Metals are first transported from the root to inside the xylem apoplast with help of different transporter proteins. This process is limited by transporter rate, substrate affinity and substrate specificity. In reality metals are accumulated differently at different levels. For example, accumulation may be found in epidermis and trichomes at the tissue level, and in vacuoles or cell walls at the cellular level (Fig. 3). Haydon and Cobbett (2007) reported that these metals are bounded by chelation proteins, which are involved in metal detoxification.

Heavy metal contamination of food is an emerging issue for food safety and quality assurance (Sharma et al., 2009). Pradhan and Kumar (2014) found that plots of arable land, 50 m, 100 m and 500 m away from a recycling site, were contaminated with Cu (23.07 mg/kg), Zn (78.18 mg/kg), Fe (106.37 mg/kg), Zn (68.48 mg/kg), Pb (0.76 mg/kg), Cd (0.049 mg/kg). Crops grown in contaminated soils can accumulate potentially harmful levels of toxic substances or heavy metals (Sharma et al., 2006, 2007). The excessive deposition of such materials in agricultural land may adversely affect normal soil processes and cause heavy metal uptake by crops, leading to deleterious effects on food quality and safety. Therefore, a human health risk is posed by dietary intake of vegetables grown in contaminated soils (Singh et al., 2010). To some degree, this high transfer factor is due to low soil pH; higher pH can stabilize soil and decrease the leaching of toxic elements (Li et al., 2004; Zheng et al., 2012).

3.6. Concentration of PBDEs and PCBs

Open land is the first destination for e-waste discarded from recycling sites after dismantling and acid processing, and the acid processing causes it to leach a variety of pollutants such as PBDEs (Luo et al., 2009a, 2009b; Tang et al., 2010b). Ma et al. (2009) reported that vegetables are affected by the processes carried out in nearby e-waste recycling sites (Chan et al., 2013; Wang et al., 2011a, 2011b, 2011c). The level of PBDEs, PCBs are shown in Table 5. This table compares the total levels of PBDEs, PCBs and other pollutants obtained from Indian e-waste processing sites. The concentrations of PBDEs and PCBs at the Mandoli industrial area and Shastri Park were higher than that of the separation workshop at Shastri Park (Brigden et al., 2005). Many studies indicate that soil contaminated with PCBs and PBDEs may have a negative impact on the natural environment (Luo et al., 2009a, 2009c; Wu et al., 2009; Zhao et al., 2008). In addition, an analysis by Ren et al. (2015) found that levels of PCDD/Fs were 3.2–31.7 pg/m^3; 0.063–0.091 pg/m^3 and 5.8–12.4 ng/kg in the air at a background site and farmland soil, respectively near an e-waste site.

4. Discussion

Improper handling and management of e-waste is one of the main causes of environmental pollution and degradation of several cities, particularly in developing countries, because of lack of regulations and appropriate treatment facilities. According to the many researchers, it can be known that the heavy metal pollution of e-waste in India has been spreaded from the informal activities to the surrounding environment (soil, air, dust and plants) (Ejiogu, 2013; Fujimori et al., 2012; Ha et al., 2009; Islam et al., 2015; Terazono et al., 2012; Zhang et al., 2012; Zhao et al., 2010). Therefore, some effective measures should be carried out to relieve the environmental pollution of heavy metals. In order to better understand the potential environmental and health risk of heavy metals pollution, a long-term risk assessment needs to be carried out on the leachability and migration potential of these toxic metals at the contaminated sites (Song and Li, 2015a, 2015b; 2014a; Liu et al., 2015; Li et al., 2015a,b).

Due to higher level of heavy metals in the informal e-waste recycling sites, especially for the two place; Mandoli Delhi and Bangalore, the engaged workers and residents are facing a potential higher exposure of these substance over the control areas. Fang et al. (2013) estimated that, during these recycling processes, dust containing heavy metals will be released into the air to impact the environment and the health of the engaged unprotected workers. And also suggested for special masks for filtering $PM_{2.5}$ are needed to mitigate the direct oral inhalation of these pollutants. Meanwhile, Zeng et al. (2015) explored the impacts of gaseous emissions and pollutant impact on environmental and human health in China (Zhang et al., 2013; Zeng, 2014).

Since 2012, India has enforced e-waste management and handling regulations established in 2011, but in realistic way, they are not implemented in a systematically, nor have been able to control the harmful activities of the informal sector. Regulation does not seem able to inhibit the illegal trans-boundary movement of e-waste and it has been suggested that India adopt the policy of neighboring countries such as China, for a systematic collection channel and recycling using environmentally sound technologies. At present more than 100 e-waste recycler/dismantlers are working in India's formal e-waste recycling sector (CPCB, 2014), but they receive only a small portion of total e-waste, because the informal

Table 4
Concentration of heavy metals in plants samples around e-waste recycling area in India described in previous reports (Pradhan and Kumar, 2014).

Location	Sampling site	Substance concentration (mg/kg)
Informal recycling site, New Delhi	Dumping site	Cu: 23.07; Pb: 0.76; Cd: 0.049; Zn: 78.18; Fe: 106.37
Informal recycling site, New Delhi	Arable land, (50 m away from recycling site	Cu: 11.43; Pb: 0.005; Cd: 0.023; Zn: 67.67; Fe: 89.49
Informal recycling site, New Delhi	Arable land, (100 m away from recycling site)	Cu: 11.24; Pb: 0.007; Cd: 0.004; Zn: 68.48; Fe: 90.32
Informal recycling site, New Delhi	Residential land, (500 m away from recycling site)	Cu: 11.08; Pb: 0.006; Cd: 0.003; Zn: 68.44; Fe: 88.47

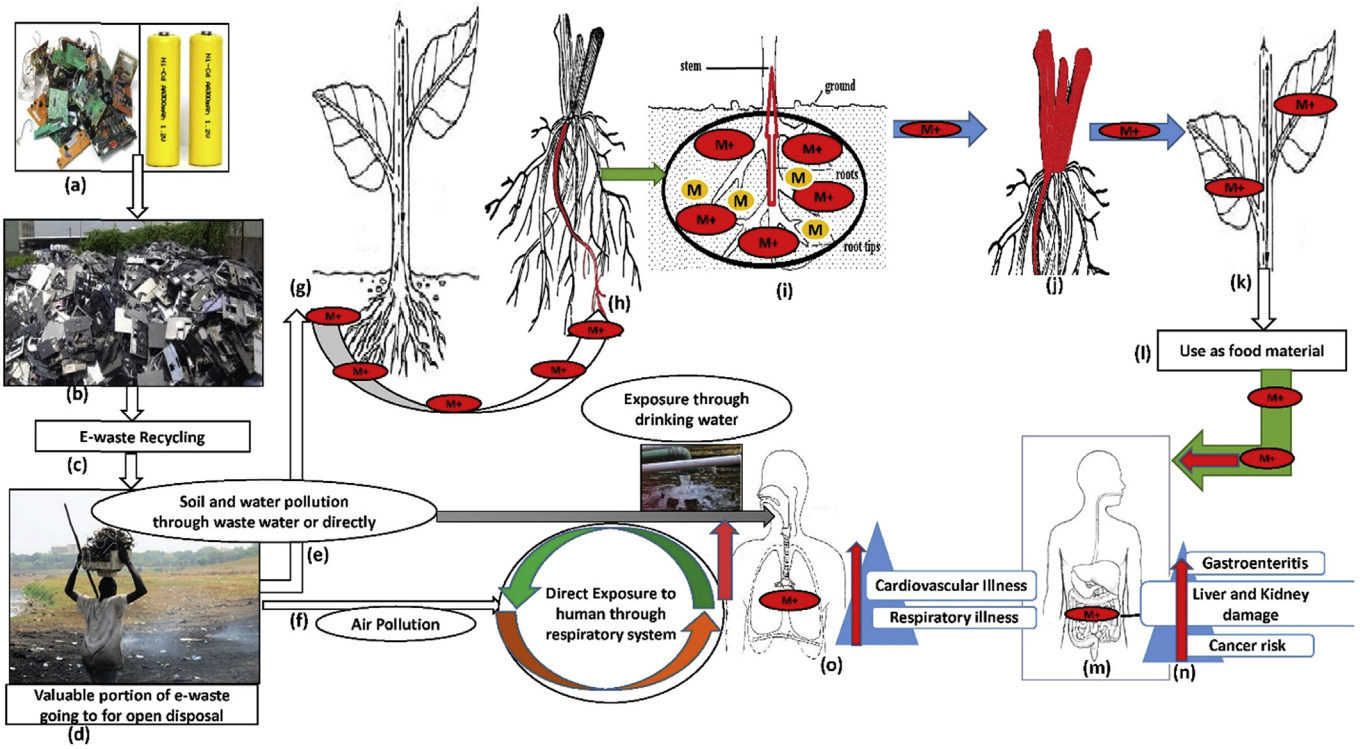


Fig. 3. Flow of e-waste contaminant via different pathways into the human body: a) Discarded e-waste; b) Storage of -waste; c) E-waste recycling, either formal or informal; d) Open disposal of parts after recycling; e-f) Release of pollutants into soil, water and air from open disposal of parts: g) Metal leaching surface soil near plants; h) Movement of metal and other substances through translocation; i) Interaction of roots, metals, and microbes; j–k) Transfer of metal inside the plant part; l) Use of plant as food material; m) Indirect movement of metals into the human body; n) Possible health problem inside human body; o) Direct pathway of contaminants into the body, and health risk.

Table 5
Concentration of PBDEs and PCBs in dust and ash samples around e-waste recycling area in India described in Brigden et al. (2005).

Location	Type of sample	Substance concentration
Zarfarabad, New Delhi	Floor dust, Separation workshop	PCB: 23; Organochloriens: 3 mg/kg
Shashtri Park, New Delhi	Floor dust, Separation workshop	PCB: 25; Organochloriens: 7; PBDEs: 2 mg/kg
Shashtri Park, New Delhi	Floor dust, solder workshop	PCB: 34; PBDEs:3 mg/kg
Mayapuri, New Delhi	Floor dust, battery workshop	PCB: 16 mg/kg
Shashtri Park, New Delhi	Street dust workshop area	PCB: 25 mg/kg
Mandoli Industrial, New Delhi	Burned fragments Acid workshop	PBDEs: 8 mg/kg
Mandoli Industrial, New Delhi	Ash, burning site	PBDEs: 23 mg/kg
Ibrahimpur, Delhi	Ash, Open burning site	PCBs: 15 mg/kg

recycling sector is so well established in Indian communities. Although valuable metals are recoverable, improper recycling processes are not only inefficient at recovering valuable metals but also are damaging our natural environment.

On the basis of data obtained from the published literature focusing on e-waste recycling worldwide, we offer an evaluation of the present situation of heavy metals contamination of surface soil, air, dust, water, effluent and plants in the India. Similar studies have represented evidence on surface soil, air, dust, water, effluent and plants in India as well as at the global scale (Bai et al., 2011; Brigden et al., 2005; Chen et al., 2010a, 2010b; Chi et al., 2011; Green Cross 2006; Ma et al., 2009; Pradhan and Kumar 2014). The surface soil, air, and ground water near e-waste recycling sites have been actively polluted with organic substances and heavy metals (Cu, Cd and Pb) (Wong et al., 2007a; Yu et al., 2006). Ha et al. (2009) proved that e-waste processing sites are highly contaminated with heavy metals. Huang et al. (2011) found that PBDEs released from e-waste processing site contaminated the land and were transferred to human beings via plant uptake in the food chain. Similarly several researchers have already reported on soil plant association (Huang et al., 2010; Zhao et al., 2008; Mueller et al., 2006). Therefore, it is urgently necessary to resolve this issue in an ecofriendly manner. Due to the limited published literature, we could not use statistical analysis.

The present investigation suggested that surface soil, air, dust, water, effluent and plants of India were seriously polluted by various heavy metals and other pollutants during e-waste processing, which might cause negative effects on people associated with these activities.

4.1. Comparative environmental pollution resulted from e-waste in some developing countries

A UNEP-based report suggested that level of metals in the environment are higher now than in the past owing to its anthropogenic activities extended inordinately such as mining (waste rock and tailing), extraction (water pollution) (Liu et al., 2014), and energy use in refining (in terms of diesel or coal fired for electricity)

reflected as atmospheric emission followed by metal recycling might be affected to environment respectively (van der Voet et al., 2013). In the both countries China and India some facilities have been built to enable the proper technology for e-waste disposal. Among these countries, China at least possesses huge quantity in such things as the smelting furnaces needed for recycling non-ferric metal. China is thus seen as the large-scale handler of e-waste that, with the help of partnering and technology transfer, has the potential for building facilities for handling a significant amount of it properly (UNEP, 2009). In Table 6, comparison of e-waste recycling and their impact among three countries India (Balde et al., 2015; Ha et al., 2009; Jha et al., 2011) China (Balde et al., 2015; Fang et al., 2013; Wang et al., 2009; Wu et al., 2015) and Nigeria (Olafisoye et al., 2013) are presented.

In fact, China also suffering due to informal e-waste recycling and in another hand they are regularly evaluated their recycling efficiency at different prospective, legal system, formal recycling systems and advanced integrated process. For instance, China has updated their implemented regulation of e-waste, by covering fourteen type such as discarded television, microcomputer, washing machine, refrigerator, and air conditioner, hood, printer, copier, fax machine, electric-water-heater, gas water heater, monitor, mobile phone, and single-machine telephone, are in the new catalogue (Liu et al., 2015; Li et al., 2015a). In additionally extensive research work has been carried out in field of e-waste covering regulation and technology level (Zeng et al., 2015; Song et al., 2015a), whereas, very limited work done in case of India.

Today's Asian countries (India, China) and African countries (Nigeria, Ghana) are first the choice for the e-waste disposal/movement from developed counties, leading to rapid environmental deterioration. We noted that the concentrations of pollutants in e-waste discard yards and recycling sites in these selected countries are potentially high. Ogungbuyi et al. (2012) estimated that 0.1 Mt was directly imported out of 0.36 Mt of e-waste is recycled using inferior standards in Nigeria. Similarly, a study conducted by Atiemo et al. (2012) tested samples from e-waste dismantling & burning sites in Accra and Ghana and found high concentrations of heavy metals (Zn, Cu, Pb and Cd) with levels of 28,957 mg/kg & 30,384 mg/kg; 16,318 mg/kg & 16,627 mg/kg; 3162 mg/kg 1321; 52.1 mg/kg 71.6 mg/kg, respectively for dismantling and burning site. However, many studies already documented the need, in China, for monitoring of environmental pollutants owing to its heavy metals and brominated flame retardants inhabitant in soil and plant (Leung et al., 2006; Shen et al., 2009; Song and Li, 2014a). These studies have proved that the informal recycling sectors are still operating in Africa with primitive methods and tools with negative consequences for the natural environment (Balde et al., 2015). These studies and reports concur that air, soil, dust, water and waste water are major sites of pollutants. The level of heavy metals and pollutants in India, China, Ghana, and Nigeria far exceeded the standard limit for levels of pollution in all countries. This scenario directly reflects the huge quantity of e-waste long processed in these countries.

4.2. Mechanism of soil contamination and an expected remedial solution

Metals can penetrate into the soil and seep into groundwater. Rahman et al. (2012) suggested that the potential amount of heavy metals are transported from the surface soil through rainfall and seepage to the ponds in the rainy season. Yuan et al. (2011) explained that, due to low level of the technology used to treat e-waste, and to the persistence of heavy metals, large amounts are transported from the soil surface via rainfall and seepage into ponds during the rainy season, contaminating the aquatic environment, and exposing nearby workers and residents to potential toxicity. It is also appears that plants growing in contaminated land straggle in terms of their growth and development, and disposed e-waste may lead to possible negative impacts on humans such as cardiovascular illness, respiratory illness, gastroenteritis and liver, kidney damage (Brigden et al., 2008). This is due to the presence of excesses concentration of heavy metals and other pollutants (Fig. 3). Therefore, the above risk may be controlled through various preventing measures such as systematic collection systems and appropriate recycling e-waste facilities. Therefore, there is an urgent need to take possible remedial approach for reforming of

Table 6
Comparison of concentration of heavy metals among e-waste recycling areas in India, China, and Nigeria described in previous reports.

Item	India[a,b,c]	China[a,d,e,f]	Nigeria[g]
E-waste generation	1.7 Mt	6.0 Mt	0.22 Mt
Formal recycling rate (%)	5	34.6	ND
Soil pollution	Dumping site: Cr: 73; Cd: 2.33; Cu: 592; Mn: 449; Pb: 297; Zn: 326 Recycling site: Cr: 54; Cd: 0.47; Pb: 126; Mn: 619; Zn: 129; Cu: 429 μg/g	Dumping site: Cd: 52; Cr: 2.51 Cu: 107; Mn: 1.01; Ni: 2.52; Pb: 111; Zn: 5 Burning site: Cd: 195; Cr: 3.45; Cu: 413; Mn: 1.12; Ni: 2.89; Pb: 115; Zn: 5.40 Acid-leaching site: Cd:146; Cr: 6.51; Cu: 621; Mn: 1.57; Ni: 23.5 Pb: 73.6 Zn: 8.82	Dumping site: Pb: 502; Cd: 7.82; Zn: 66.9; Cr: 32.65; Ni: 84.24
Water pollution	Waste solution: Al: 1315; Cd < 1; Cu: 185; Ni: 9; Pb: 4; Zn: 17 ppm	Well: Cd: 5.60; Cr: 0.058; Cu: 112; Mn: 138; Ni: 3.07; Pb: 1.37	Well: Pb: 1.8; Cd: 0.006; Zn: 0.84; Cr: 0.25; Ni: 1.23
Air pollution	Recycling site: Cr: 18; Mn: 59.6; Cu: 111; Zn: 191; Mo: 81.6 ng/m3	Mechanical workshop: Cr: 0.554; Cu:27.76; Cd: 0.108; Pb: 12.34 mg/g; Manual workshop: Cr:0.436; Cu: 31.80; Cd: 0.398; Pb: 2.043 mg/g	ND
Human health	Cr: 0.29; Mn: 1.16; Cu: 23; Zn: 141; Mo: 0.041; Ag: 2.1 μg/g	As: 0.282; Cd: 0.209; Cr: 1.16; Cu: 10.2; Mn: 1.03; Ni: 0.812; Pb: 2.98 mg/g	ND

ND: No Data.
[a] Balde et al. (2015)
[b] Ha et al. (2009).
[c] Jha et al. (2011).
[d] Wu et al. (2015).
[e] Wang et al. (2009).
[f] Fang et al. (2013).
[g] Olafisoye et al. (2013).

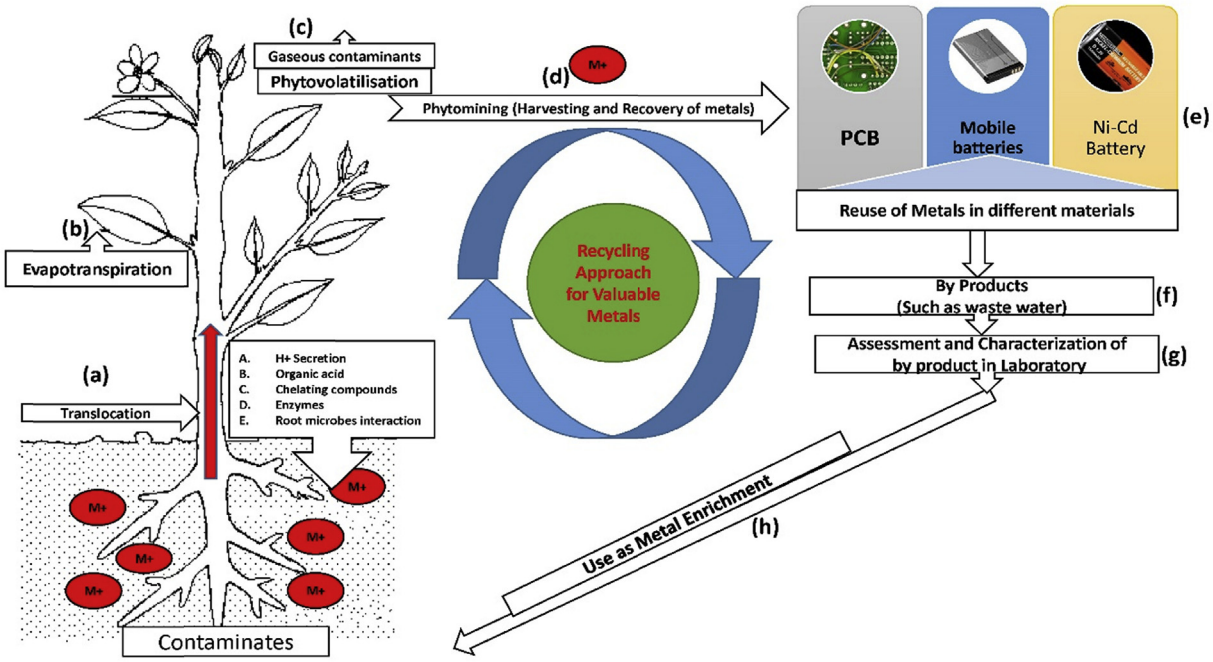


Fig. 4. Remedial approach for metals contaminated soil and recycling; a) Translocation; b) Evapotranspiration; c) Phytovolatilisation; d) Phytomining; e) Reuse of metals; f) Product collection after manufacturing; g) Laboratory examination; h) If under standard permissible limit, then use as metal enrichment.

contaminated sites. The soil is an excellent habitat for variety of microbes associated with plants, and are also involved in phytoremediation process such as stabilization, degradation in rhizosphere and plant, accumulation inside tissue and volatilization (Liu et al., 2015; Tang et al., 2013; Zhang et al., 2010). Specially, soil fungi have great tolerance ability such as metal adsorption on cell wall surface (also known as biosorption), bioaccumulation (transportation and cellular incorporation) and redox or methylation reactions (also called transformation) (Gadd, 1986, 1993). Many studies have systematically described the response of arbuscular mycorrhizal fungi (AMF) to metals (Huang and Cunningham, 1996; Moynahan et al., 2002; Meier et al., 2011, 2015; Barea et al., 2013; Seguel et al., 2013). Rajkumar et al. (2012) suggested that the microbes produce several kind of extracellular polymeric substances (EPS), mucopolysaccarides and proteins that play a major role in toxic metals complexes which slow its mobility rate in the soil (Seguel et al., 2015). Many publications proved that microbes and plants can be used for remediation of these contaminated land (Chen et al., 2015; Bizkarguenaga et al., 2015; Song et al., 2015b; Ni et al., 2014; Chen et al., 2010a, 2010b).

This paper is principally focused on India, and is very similar to a study carried out by Song and Li (2014a) but we have taken an extensive approach to solve this problem with phytoremedial technology (Fig. 4). The metal and PBDEs, PCBs and organochlorine in air, water, soil and dust were differently distributed at different sites, but we used only the mean concentration of pollutants. Due to certain limitation it was not feasible for us to apply the statistical methods. However, direct exposure might be prevented through regular use of mask and gloves (Song and Li, 2014a). Our study also covered additional pollutants, not covered by Song and Li (2014a).

5. Conclusions

This paper explores the environmental pollution from e-waste recycling at many small formal and informal workshops in India. The traditional e-waste processing through improper channels in India has resulted in the huge quantity of heavy metal and other pollutants into the natural environment which has a negative impact on natural ecosystems (soil, water, dust and plant). Therefore, this study aims to give a clear picture of environmental pollution from e-waste processing by an informal sector well established in many Indian communities. According to this systematic approach, the informal sector should be integrated with the formal sector at collection channels, regular monitoring should be carried out and e-waste recycling facilities should be handled in a preventive manner. The contaminated soil can be remediated through integrated phytoremedial and microbial systems, followed by use of by-products as metal enrichments to soil. Further research is needed for better understanding of long-term impact of substance and integrated plant-microbial system application in sustainable management.

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Abbreviation

BFRs	brominated flame retardants
E-waste	electronic waste
EPS	extracellular polymeric substances
MoEFCC	Ministry of Environment, Forestry and Climate Change
PAHs	polycyclic aromatic hydrocarbons
PBDEs	polychlorinated diphenyl ethers
PCBs	polychlorinated biphenyls
PCDD/Fs	polychlorinated dibenzo-p-dioxins and dibenzofurans
UNEP	United Nations Environment Programme

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